

continuation

Continuation Document

Revision: 86 **Last modified:** 2026-06-01T23:43:49Z **Description:** Sacred invariant resumption document for Helix Cluster OS **Authority:** Constitution §12.10
Maintainer: Operator + AI loop

§0: How to use this document

Any CLI agent resuming work on this project MUST read this file first.

§1: Recently Completed Work (last 10 commits)

Commit	Message
02cd05a	Foundation wave 25: mark 3 items Completed (HXC-1254/1255/1267) — parallel TestRunner / session test FSM / helix-test CLI (verified already complete). Mutation spot-checks confirm bites. Registry 188->191 Completed / 448 Queued.
aba565b	Foundation wave 25: 2 new test-orchestration pkgs + 1 verified-already-done. HXC-1254 parallel TestRunner (NEW pkg/testing/runner: Suite registration + bounded-worker-pool parallel execution + result collection/aggregate Report sorted deterministically; barrier-proof concurrency test (serial runner deadlocks under a ctx-deadline guard), atomic peak counter proves MaxConcurrency cap, result counts incl Err->Failed (defense-in-depth: both guards must be removed to break the want-3-got-4 bite — mutation-verified), Register dup/empty-name errors). HXC-1255 session test FSM (NEW pkg/testing/sessionfsm: Idle->Setup->Running->ChaosInject->Verify->Done + any-nonterminal->Failed; legal-transition table, IllegalTransitionError leaves state unchanged, per-state callbacks fire in order, History(); mutation guard-always-allow -> illegal transition test FAILS). HXC-1267 helix-test CLI VERIFIED already complete (cmd/helix-test dst/chaos/device/snapshot subcommands wired to real pkg/testing packages w/ 13 mutation-sensitive tests; device subcmd lists real T1-T8) — verified+marked, not rebuilt. ZERO new deps, ZERO edits to existing files. Gate: build/vet/vet-integration clean, full -short -race green, dataplane ok, tmux clean (W23 fix holds); mutation spot-checks: FSM guard bites, TestRunner Err->Failed bites (both guards), cap-peak<=2 + barrier verified by review.
ace79d4	Foundation wave 24: mark HXC-1162 Completed — E2E multi-node session CRDT convergence (5 replicas, arbitrary-order gossip, converge to identical

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	Fingerprint + canonical truth; mutation-verified bite). Registry 187->188 Completed / 451 Queued.
8e8cebb	Foundation wave 24: HXC-1162 E2E multi-node session CRDT convergence (pkg/session/convergence_hxc1162_test.go, test-only, ZERO edits). Proves STRONG EVENTUAL CONSISTENCY beyond the existing 2-replica tests: 5 independent replicas each seeded with a DISJOINT partition of 100 ops, gossiped via seeded-random ARBITRARY-ORDER pairwise merges (dst = dst.Clone().Merge(src)) until 3-sweep quiescence; asserts (a) ALL 5 replicas reach an identical Fingerprint AND (b) that fingerprint equals a canonical truth replica that applied all 100 ops sequentially — over 20 seeds. Reuses existing genUpdates/replicaFrom/buildUpdate + CRDTSessionState Merge/Clone/Fingerprint. Builds on W22 pkg/checkpoint_merge: this is the convergence guarantee that makes coordinator-free session migration safe. Gate: build/vet/vet-integration clean, full -short -race green (zero failures this run), dataplane ok, tmux ls clean (W23 leak fix holds); INDEPENDENT MUTATION SPOT-CHECK: order-dependent equal-version pane tiebreak (paneWins 'return false') -> replicas DIVERGE -> 'CONVERGENCE FAILED replica 0 and 1 diverged' (the N-node convergence bite is genuinely real). ZERO new deps.
06ca1a2	Foundation wave 23: fix tmux-session test leak (CLAUDE-1/§7.1 test hygiene; PTY-exhaustion guard, W15/W22 lesson). Real-tmux-backed tests now leave ZERO leaked sessions. ROOT CAUSE was cmd/helix-session/main_test.go: TestRunServesRealRPC + TestRunGracefulShutdownStopsServing call the production run() (real tmux backend since W15) and CreateSession 2 real tmux sessions (server-assigned session- id == tmux session name) with no cleanup -> leaked on every full-suite run. FIX: added killSession(t,id) helper t.Cleanup-ing 'tmux kill-session -t ' after each CreateSession. Also pkg/session/backends/package_test.go (W23 workflow): added t.Cleanup kills to TestTmuxBackend_CreateAndKill/Lifecycle (defended against mid-test-failure leaks). VERIFIED: full -short -race ./... then 'tmux ls' => no server running (zero leaks); real-tmux proofs intact (hxc1136 exit gate untouched). Test-only changes, ZERO production code modified. NOTE flakes: internal/advisory TestLockBlocksUntilAvailable + test/e2e TestClusterLifecycleSuite FAIL only under heavy concurrent full-suite -race load, PASS isolated (load-induced timing flakes, not regressions).
639e479	Foundation wave 22: mark 2 items Completed (HXC-1152/1160) — container-checkpoint migration CRDT merge / tmux full functional Attach (send-keys+capture-pane, real marker round-trip). Mutation spot-checks confirm bites. Registry 185->187 Completed / 452 Queued.
1d79b5c	Foundation wave 22: 2 disjoint streams (both approved no-fix; mutation spot-checks bite). HXC-1152 container-checkpoint migration CRDT merge (NEW pkg/checkpoint_merge: Apply(local,checkpoint) restores+merges incoming CRDT checkpoint into a clone of local via existing CRDTSessionState.Merge LWW, Migrate(source,local) models export-A->merge-B; NEW pkg/session/crdt_checkpoint.go v2 checkpoint helper avoids import cycle; convergence tests assert EXACT LWW winner values + commutativity-by-Fingerprint + idempotence + corrupt-checkpoint sentinel; mutation skip-merge -> LWW winner reverts to wrong value FAILS). HXC-1160 tmux full functional Attach (pkg/session/backends/tmux_attach.go + tmux.go Attach: replaced the dead orphaned-io.Pipe stream with a REAL stream — Write delivers keystrokes via tmux send-keys -l/Enter, Read returns REAL pane content via tmux capture-pane -p; marker round-trip test against real tmux with bounded 20x100ms poll,

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	unique names + defer Kill, ZERO leaks; mutation neuter send-keys -> marker absent FAILS). ZERO new deps. Gate: build/vet/vet-integration clean, full -short -race green, dataplane ok; both mutation spot-checks bite; W22 tmux test verified zero-leak in isolation. NOTE pre-existing leak: pkg/session.Manager-based tests (manager.go:106 session- naming, e.g. hxc1136) can leave tmux sessions — hygiene item, not introduced here. Foundation wave 21: mark 3 items Completed (HXC-1130/1249/1243) — screen+zellij session backends (real screen, zellij ErrUnsupported) / emergency-stop+auto-recovery / 8 network fault injectors (verified already complete). Mutation spot-check confirms emergency-stop bite. Registry 182->185 Completed / 454 Queued. Foundation wave 21: 2 impl streams + 1 verified-already-done. HXC-1130 screen+zellij session backends (pkg/session/backends/screen.go real exec of /usr/bin/screen: Create -dmS / List -ls parse / Kill -X quit / SendInput -X stuff / CaptureOutput -X hardcopy; REAL integration test creates+lists(sink-side proof)+kills a uniquely-named session w/ defer-Kill, ZERO leaked sessions verified; zellij.go returns typed ErrBackendUnavailable since zellij absent — no fake; orchestrator fixed the Attach stream to deliver Write as REAL keystrokes via SendInput instead of an orphaned io.Pipe). HXC-1249 emergency-stop + auto-recovery (pkg/testing/scenario/stop.go: StopController.EmergencyStop snapshots ActiveFaults + RestoreAll ->
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15977c0	StopResult{FaultsRecovered,HaltLatencyMS,WithinBudget,RecoveredNames}; halt latency modelled as data (faults*perFaultCost), NO real sleep; AutoRecover asserts no faults remain; post-stop ActiveFaults()==0 + WithinBudget false-case are the bites). HXC-1243 8 network fault injectors VERIFIED already complete (NetworkPartition/PacketLoss/LatencyInjection + PacketReorder/PacketDuplication/BandwidthThrottle/AsymmetricPartition/DNSBlackhole, all with passing tests) — verified+marked, not rebuilt. ZERO new deps. Gate: build/vet/vet-integration clean, full -short -race green, dataplane ok; mutation spot-check: B skip-RestoreAll -> post-stop ActiveFaults!=0 FAILS (bite real); A real screen create/list/kill verified sink-side, zero leaks. Foundation wave 20: mark 2 items Completed (HXC-1248/1244) — YAML chaos scenario engine (blast-radius+abort) / simulated node fault injectors. Plus HXC-1110 /metrics coverage completed for the 6 gRPC-only services. Mutation spot-checks confirm bites. Registry 180->182 Completed / 457 Queued.
bc99718	

§2: Environment Snapshot

Property	Value
Branch	main
Commit	02cd05a
Timestamp	2026-06-01T23:43:49Z

§3: Active Work

HXC	Title	Status
HXC-922	Node service, advisory locks, chaos tests, docs	✓ Done
HXC-921	Web UI, K8s/Helm, integration, E2E, benchmarks	✓ Done
HXC-920	Health, security, build, wireguard services	✓ Done
HXC-919	htmux, messaging, GPU, WASM, storage, build manifest	✓ Done
HXC-918	Wire scheduler/session gRPC to real backends, etcd discovery	✓ Done
HXC-916	Stub expansion (config, retry, ratelimit, validator, build cache)	✓ Done
HXC-914	LLM, policy, setup, distributed lock	✓ Done
HXC-913	Health, security, ClassAds parser	✓ Done
HXC-912	Gateway, helixd, helix-agent	✓ Done
HXC-911	Scheduler, Session gRPC services	✓ Done
HXC-910	Console, security, testing infra	✓ Done

§4: Next Planned Work

All MVP, Phase 2, Phase 3, and Phase 4 are **COMPLETE**. Potential future enhancements:

1. **Performance optimization** — Profile and optimize hot paths
2. **Multi-region support** — Cross-region cluster federation
3. **GPU scheduling v2** — Multi-GPU, fractional GPU allocation
4. **Web UI real-time** — WebSocket integration for live updates
5. **Operator pattern** — Kubernetes operator for cluster management

§5: Known Issues / Blockers

- None

§6: Quick Commands

Run all tests (requires GOWORK due to workspace modules)

```
GOWORK=$(pwd)/go.work go test ./... -race -count=1
```

Build all binaries

```
GOWORK=$(pwd)/go.work go build ./cmd/...
```

Build web UI

```
cd web && npm run build
```

Render Helm chart

```
helm template helix-cluster deploy/helm/
```

Run benchmarks

```
go test -bench=. ./test/benchmark
```

Run chaos tests

```
go test -race -count=1 ./test/chaos
```